

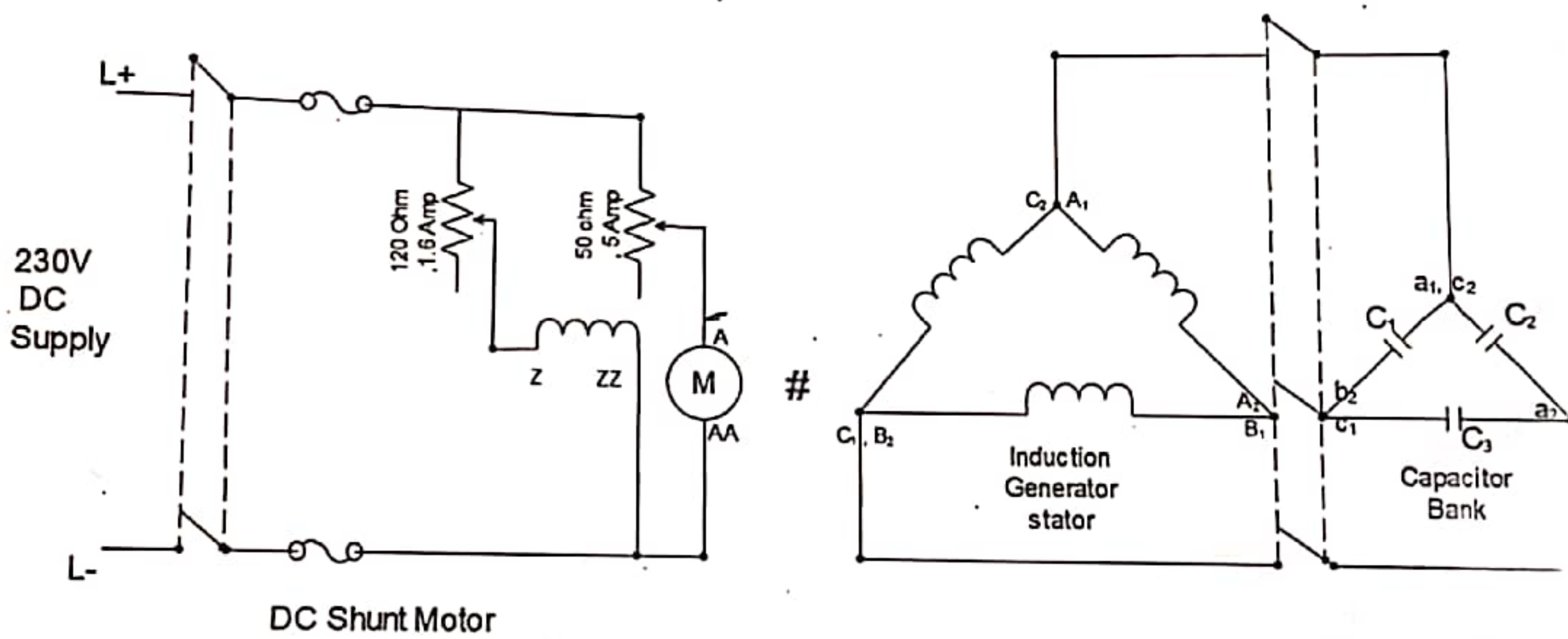
Aim:

Experiment No- 8

Apparatus Required:

S. NO.	INSTRUMENTS	SPECIFICATIONS	QUANTITY
1.	RHEOSTAT	50Ω , 5A	1
2.	RHEOSTAT	120Ω , 1.6A	1
3.	CAPACITOR	36μF , +- 15%	3
4.	VOLTMETER	0-600V	1
5.	TACHOMETER		1

Circuit Diagram :



Page No _____

Theory:

Induction motor can be made to work as a generator if we rotate its rotor by a prime mover at a speed greater than N_s . In this experiment we use DC shunt motor as a prime mover excited by a DC supply.

$$N_s = 120 \cdot f / P \quad \text{where } f : \text{input frequency}$$

P : pole per phase

N_s : synchronous speed

The above synchronous speed across the stator winding of the induction emf is very low so as to increase this induced emf we connect a capacitor bank across the stator winding, by connecting the capacitor the rotor speed is decreased as power factor increases.

Procedure:

1. Make connection as shown in figure.
2. Now cut out the armature rheostat as it is initially placed at a full load resistance.
3. Now increase field resistance, initially it is kept at low resistance so that the excitor flux (ϕ) decreases and dc shunt motor rotates above the synchronous speed $N(s)$

$$\uparrow N \propto \frac{E_b}{\phi \downarrow}$$
4. The dc shunt motor is mechanically coupled to induction motor which also rotates with dc motor the speed of both side is same.
5. Now measure the emf across the rotor winding by connecting capacitor and without capacitor for successive reading, now measure the emf across the rotor winding with connecting capacitor and without capacitor.
6. For successive reading change the additional resistance and obtain different speed at corresponding reduced voltage.
7. Now shut down the dc motor.

Observation Table :

S.NO.	WITH CAPACITOR		WITHOUT CAPACITOR	
	VOLTAGE (V)	SPEED (RPM)	VOLTAGE (V)	SPEED (RPM)
1.				
2.				
3.				
4.				

Result:

The obtained readings show that for the given self excited generator voltage developed and speed increases and decreases respectively on adding capacitor bank .

Precautions:

1. Do not touch any terminal or switch without ensuring that it is dead .
2. Never work alone in the laboratory.
3. Make sure that all the connections are right, before switching on any circuit , faulty connections may cause drawing large power resulting in damage of parts of instruments .
4. The circuit should be de-energized before changing any connections